

LISA NGUYEN

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EDUCATION

University of Michigan

PhD in Applied Physics and Scientific Computing

Topic: Spatio-Temporal Dynamics and Causal Drivers of Hydro-climatic Extremes Under Climate Change

Expected Graduation: Dec 2026

University of California Berkeley

B.A. in Physics, Major GPA 3.833

B.A. in Pure Mathematics, Major GPA 3.882

Graduation: Dec 2019

AWARDS/FELLOWSHIPS/PROGRAMS

- Rackham Predoctoral Fellowship (≈ 85 selected campus-wide)
- University of Michigan student delegate for COP30 (1 of 8 candidates selected campus-wide)
- Thriving Earth Exchange Community Science Fellowship
- Future of Research in Climate, Earth, and Energy (FORCEE) DoE Internship
- Rackham Merit Fellowship
- Michigan Institute for Computation Discovery (MICDE) & Engineering PhD

RESEARCH EXPERIENCE

Drought Synchronicity over the Contiguous United States

Authors: Nguyen, L., & Ombadi, M.

PhD research: UMich

Apr 2025-Dec 2025

- Developed an integrated analytic framework merging Event Coincidence Analysis, wavelet analysis, and ML-based feature attribution (Random Forest + SHAP) to quantify drought co-occurrence, propagation pathways, and large-scale synchronicity patterns across the U.S.

Record Breaking Extreme Precipitation Events

Authors: Nguyen, L., Todorovic, L., & Ombadi, M.

PhD research: UMich

Apr 2024 - present

- Mapped the emergence of unprecedented precipitation extremes under global warming in CMIP6 models and integrated socioeconomic datasets to compute global population exposure and vulnerability patterns.

Future Of Research in Climate, Earth and Energy (FORCEE) Intern Sandia National Laboratories

Supervisor: Lauren Wheeler

May 2023 - Oct 2023

- Validated a modified build of the DoE's earth system model, E3SMv2, carrying a more complete representation of stratospheric chemistry, verifying model behavior before production simulations.
- Configured and ran E3SMv2 simulations of stratospheric aerosol injections (SAIs), implementing a proportional-integral feedback controller adapted from [dan-visioni](#). The controller adjusted injection rate at specified locations while simultaneously controlling three temperature metrics.

Stratospheric Investigation of E3SMv2

Advisor: Christiane Jablonowski

PhD research: DoE and UMich

May 2022 - Oct 2023

- Investigated stratospheric tropical circulation in the DoE's newest earth system model, E3SM version 2 (E3SMv2), comparing the global climate model to reanalysis data.
- Quantified the momentum transport through Transformed Eulerian Means to evaluate the Brewer Dobson Cycle. Found a significant dry bias of water vapor in the model as well as an increase in the stratospheric "Tape Recorder" propagation speed.

Frequency Analysis of High Intensity Precipitation

Advisor: Brian Arbic

PhD research: UMich

Aug 2021 - Jan 2021

- Using local, but high frequency rain gauge data from NOAA to compare against high resolution climate models. Compared the frequency space of models vs observations by analyzing the power spectral density of rain events using data from minutes to decades.

PEER-REVIEWED PUBLICATIONS

- **Nguyen, L.** & Ombadi, M. (2026) *Characterizing Patterns of Drought Synchronicity in the Contiguous United States*. Water Resources Research. <https://doi.org/10.1029/2025WR041240>
- **Nguyen, L.**, Todorovic, L., & Ombadi, M. (2026) *Anthropogenic Global Warming Increases the Risk of Record-breaking Extreme Precipitation Events in low-income Countries*. Nature Communications Earth & Environment. <https://doi.org/10.1038/s43247-026-03649-y>
- Ombadi, M., **Nguyen, L.**, Gaur, S., Gronewold, A., & Reich, P. (2026) *Increased Dependency on Extreme Precipitation in a Warmer Climate*. Water Resources Research. <https://doi.org/10.1029/2025WR042627>

COMMUNITY LEADERSHIP

Climate Blue at University of Michigan

COP30 Student Delegate

Ann Arbor, MI

May 2025 - present

- Represented Umich as a Climate Blue delegate to the UNFCCC COP 30, producing accessible briefings and outreach materials to connect global climate negotiations with students, researchers, and local communities.

AGU: Thriving Earth Exchange

Community Science Fellow

Volunteer Community Scientists

Ann Arbor, MI

May 2025 - present

Aug 2024 - Dec 2025

- Investigated illegal dumping at Scott Memorial UMC, supported grant applications for land-use changes, led community recycling workshops, and coordinated hiring of local muralists to promote neighborhood engagement.
- Managed a climate-health initiative using LiDAR, climate, and socioeconomic data to guide equitable urban heat-resilience planning in Boynton Beach, Florida, including cooling-center siting and adaptation strategies for vulnerable residents.

University of Michigan: CLaSP GUSTO

DEI Officer and Volunteer Graduate Mentor

Ann Arbor, MI

Aug 2024 - present

- Organized DEI related activities and mentorship within the department.
- Volunteered to mentor graduate students interested as part of the GUSTO mentorship program.

Ypsilanti Math Corp at University of Michigan

Course Instructor

Math Corp Activities/HW Volunteer

Ann Arbor, MI

Summer 2024

May 2019 - Dec 2024

- Developed and implemented "Ice Cream" class curriculum on graph theory for 8th graders and high school to teach fun topics in math during the summer camp.
- Co-organized the activities room to increase awareness and engagement with mathematics for Ypsilanti children grades 7-9 at the Ypsilanti Math Corp summer camp and Super Saturdays.

FATE Give Merit Program

Mentor

Ann Arbor, MI

Sep 2022 - Dec 2024

- Mentored high school students weekly in Detroit, working on strengthening their professional skills.

University of Michigan: Applied Physics Department

Volunteer Graduate Mentor

Ann Arbor, MI

Aug 2022 - present

- Mentored incoming graduate student, Shriya Sinha. Set up weekly HW sessions and monthly dinners.

Physics Diversity, Equity, and Inclusion Committee

Outreach Committee Volunteer

Ann Arbor, MI

Aug 2021 - present

- Volunteered for outreach opportunities with the local communities. (WISE middle school demos, etc)

Michigan Math Circle

Volunteer Assistant

Ann Arbor, MI

Aug 2022 - May 2023

- Guided elementary to high school students in engaging and complex math topics.
- Gave interactive lectures and activities on accessible, but interesting math problems

Applied Physics Student Council

Authorized Signer

Ann Arbor, MI

Aug 2022 - present

- Responsible for activities hosted by the Applied Physics Department's graduate student council.

University of Michigan: College of Engineering

Volunteer Graduate Mentor

Ann Arbor, MI

Aug 2021 - Aug 2022

- Volunteered to mentor undergraduate students interested in climate and space. Set up regular HW sessions and talks to discuss various aspects of academic and social life.

Cal Women's Rugby Team

Academic Officer/(Player Position: Prop)

Berkeley, CA

Aug 2018 - Jun 2019

- Monitored and created individual academic plans, with an emphasis on fostering time management skills. Scheduled and organized study sessions and events to promote an academic lifestyle.

Mathematics Undergraduate Student Associations (MUSA)

Berkeley Undergraduate Mathmentor Program (BUMP) Mentor

Berkeley, CA

Aug 2018 - Jun 2019

- Each semester, mentored two underclass students interested in mathematics in order to foster enthusiasm for continuing math education. Met twice a week to talk about school/hobbies/life.

POSTERS/PRESENTATIONS

- **Nguyen, L.** (March 17, 2026) Invited speaker for *Student Stories from COP30: COP30 Debrief with Environmental Policy Association*, University of Michigan Environmental Policy Association
- **Nguyen, L.** & Ombadi, M. (January 29, 2026), *Quantifying Global Synchronization of Droughts in Breadbasket Regions Under a Warming Climate*, American Meteorological Society (AMS) Annual Meeting, Abstract 469617
- **Nguyen, L.** & Ombadi, M. (December 19, 2025), *Characterizing Drought Synchronicity Across the Contiguous United States (1980-2021)*, American Geophysical Union (AGU) 2025, Abstract 1969758: NH51E-0228
- **Nguyen, L.**, Li, Q., & Ombadi, M. (December 18, 2025), *Decoding Drought Dynamics: A Multi-Method Intercomparison for Event Characterization Across Spatio-Temporal Scales*, American Geophysical Union (AGU) 2025, Abstract 1969758: H41M-1332
- Todorovic, L., **Nguyen, L.**, & Ombadi, M. (December 17, 2025), *Disproportionate Increase of Unprecedented Rainfall Events in Low Income Countries*, American Geophysical Union (AGU) 2025, Abstract A31E-2108
- Gaur, S., **Nguyen, L.**, Gronewold, A., Reich, P., & Ombadi, M. (December 15, 2025), *Changes in Extreme and Annual Precipitation under Global Warming*, American Geophysical Union (AGU) 2025, Abstract GC11G-0662
- Zhong, Y., Li, Q., **Nguyen, L.**, & Ombadi, M. (December 18, 2025), *Examining the Impact of Drought on Vegetation in the Contiguous United States*, American Geophysical Union (AGU) 2025, Abstract B41I-1970
- **Nguyen, L.** (August 13, 2025) Invited speaker for *Webinar: Science with Purpose: How Scientists Can Help Empower Community Solutions*, American Geophysical Union, Thriving Earth Exchange
- **Nguyen, L.**, Ombadi, M. (2024), *Quantifying Global Event Synchronization of Droughts in a Warming Climate*, American Geophysical Union (AGU) 2024, Abstract 1620436: GC33G-0234
- Jablonowski, C., Hollowed, J., **Nguyen, L.**, Hughes, O., Ehrmann, T., Wagman, B., & Hillman, B. (2023), *A Closer Look at E3SMs Stratosphere: Circulation Biases, Transport, Tracers*, DoE CLDERA All-Hands Meeting, Albuquerque, NM, USA, Oct 17-19, 2023
- Jablonowski, C., Hollowed, J., **Nguyen, L.**, Ehrmann, T., Wagman, B., & Hillman, B. (2023), *Tropical Stratosphere-Troposphere Interactions in Selected CMIP6 Models*, Joint SPARC DynVar - SNAP Meeting, Munich, Germany, Oct. 9-13, 2023
- Jablonowski, C., **Nguyen, L.**, Kinnison, D., & the CLDERA project team (2023), *Characteristics of the stratospheric tropical circulation in DoEs Energy Exascale Earth System Model E3SMv2*, CESM Workshop 2023, Boulder, CO, USA, June 12-14, 2023
- Jablonowski, C., **Nguyen, L.**, Golaz, J.-C., Rosenbloom, N., & Meehl, G. A. (2023), *Characteristics of the Stratospheric Tropical Circulation of the Energy Exascale Earth System Model E3SMv2*, European Geosciences Union (EGU) 2023, Abstract EGU23-15173

TEACHING

- **Physics 141: Elementary Lab** (GSI) - Winter 2024

WORK EXPERIENCE

RSAE Labs

Bookkeeper

Tampa Bay, FL

Jul 2021 - present

- In charge of payroll, accounts receivable, accounts payable, and financial reports. Streamlined several administrative processes, to facilitate transparency.

Omnium

Data Scientist

San Francisco, CA

Jul 2019 - Aug 2021

- Classification of SKU rationalization using various clustering models.
- ETL on Consumer Package Goods (CPG) consumption data to derive key insights for client business intelligence.
- Optimized pivotal client business metrics (both top and bottom line) given budget constraints by building a heuristic algorithm to a Mixed Integer Non-Linear programming problem.
- Restructured and reformatted internal data capture processes and automated lower level analysis by establishing a coherent data model and pipeline using AWS and Power BI.
- Significantly streamlined and implemented intern training to create a systematic transition from training to internal and client facing company projects. Served as a project mentor for 8 different interns.

Sandia National Laboratories

Nuclear Radiation and Detection Intern (NSSC affiliate)

Livermore, CA

Aug 2018 - Jun 2019

- Oversaw and implemented a methodical study of neutron production in high-Z materials (lead and tungsten) on a per-muon basis, with the goal of calculating the neutron multiplicity distribution for both spallation and capture channels. Optimized the data acquisition system, independently researched and ran experiments.